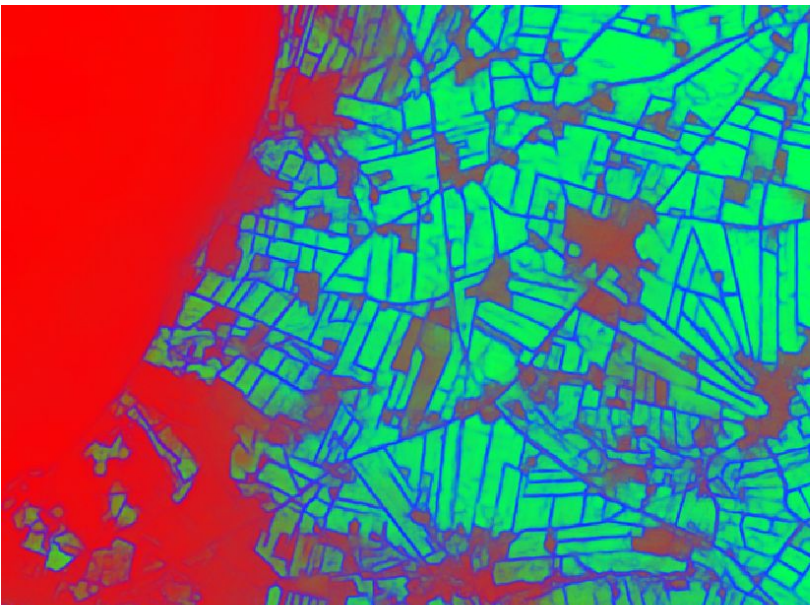


Core-collapse Supernovae in the Dark Energy Survey

Matthew Grayling, Claudia Gutierrez, Mark Sullivan

A brief aside...



Introduction

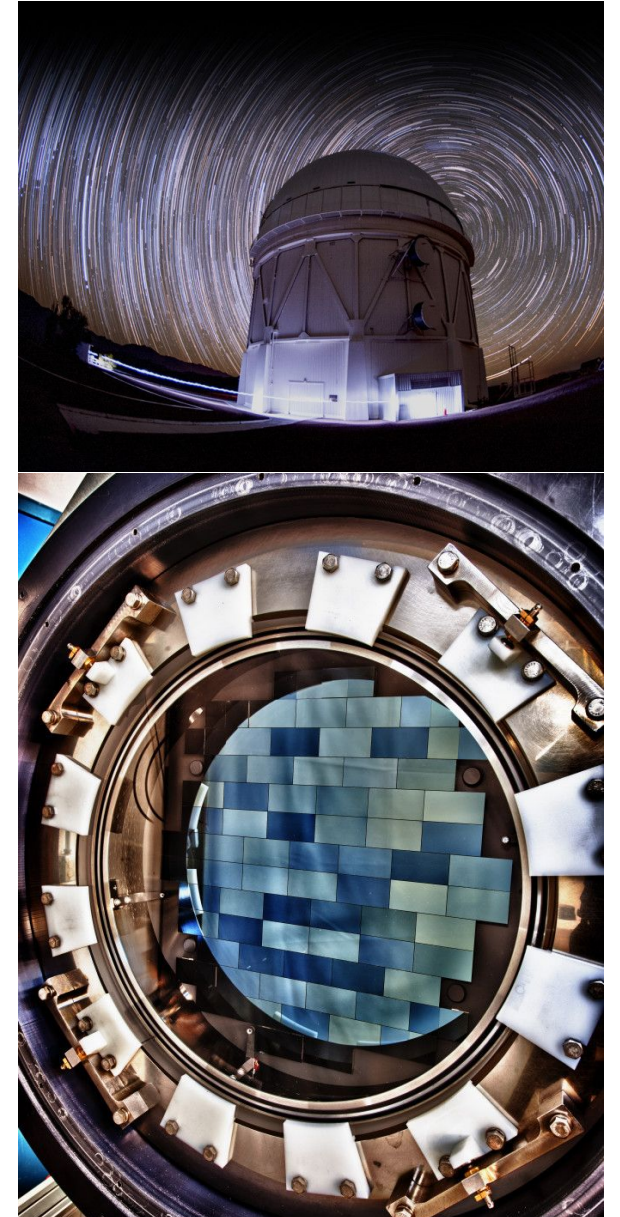
- Core-collapse supernovae (CCSNe): most common type of cosmic explosion, but not well understood
- Their distribution in luminosity - the 'luminosity function' - is the most basic population diagnostic
- Having accurate luminosity functions is important for a variety of reasons:
 - Helps constrain CCSNe simulations
 - Critical for accurately simulating supernova surveys for rate calculations, investigating CCSNe contamination of Ia samples etc.

Previous Work

- Two main previous studies looking at CCSNe luminosity functions
- Li et al. 2011
 - Used Lick Observatory Supernova Search, 105 CCSNe
- Richardson et al. 2014
 - Based on full Asiago Supernova Catalogue
 - No uncertainties, for now we have not compared to this

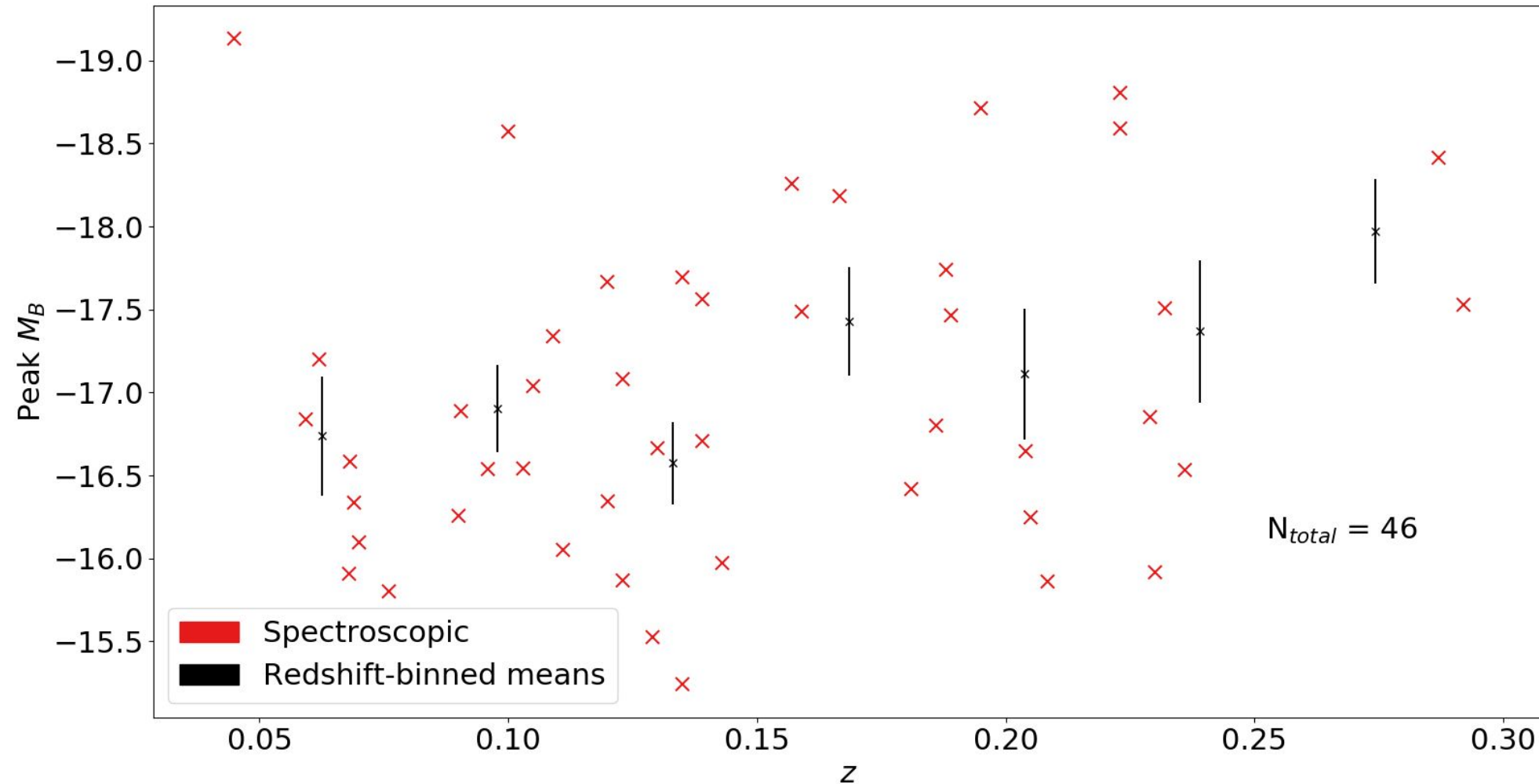
Dark Energy Survey (DES)

- Aim - “Everything with one survey”, studying dark energy using multiple probes
- DES SN
 - Observed 30 square degrees in four filters every 7 nights for 5 years
 - Very deep survey, ~ 24 mag
 - Doesn't pick up many local objects but very good for detecting objects at high redshift

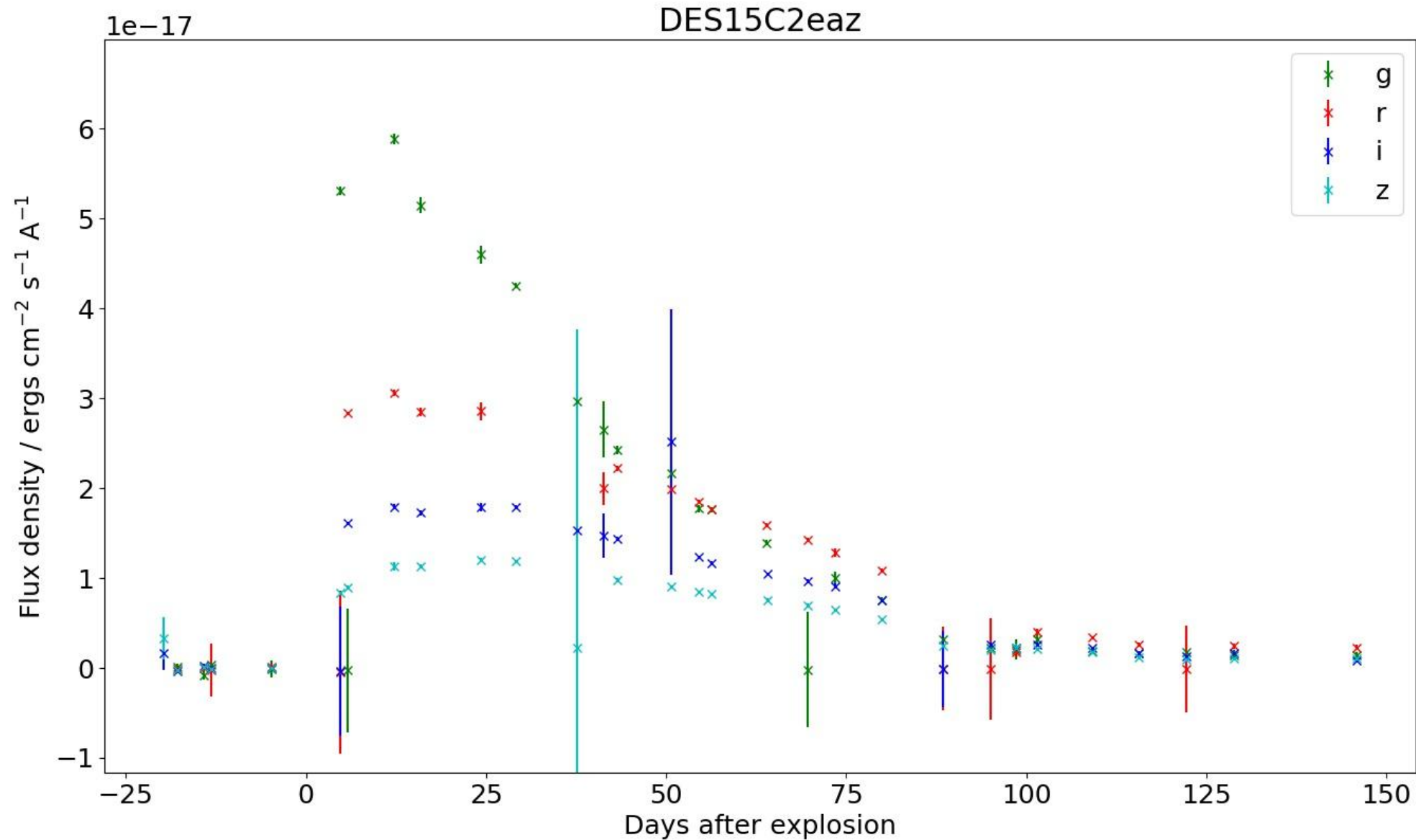


DES Sample - Spectroscopic

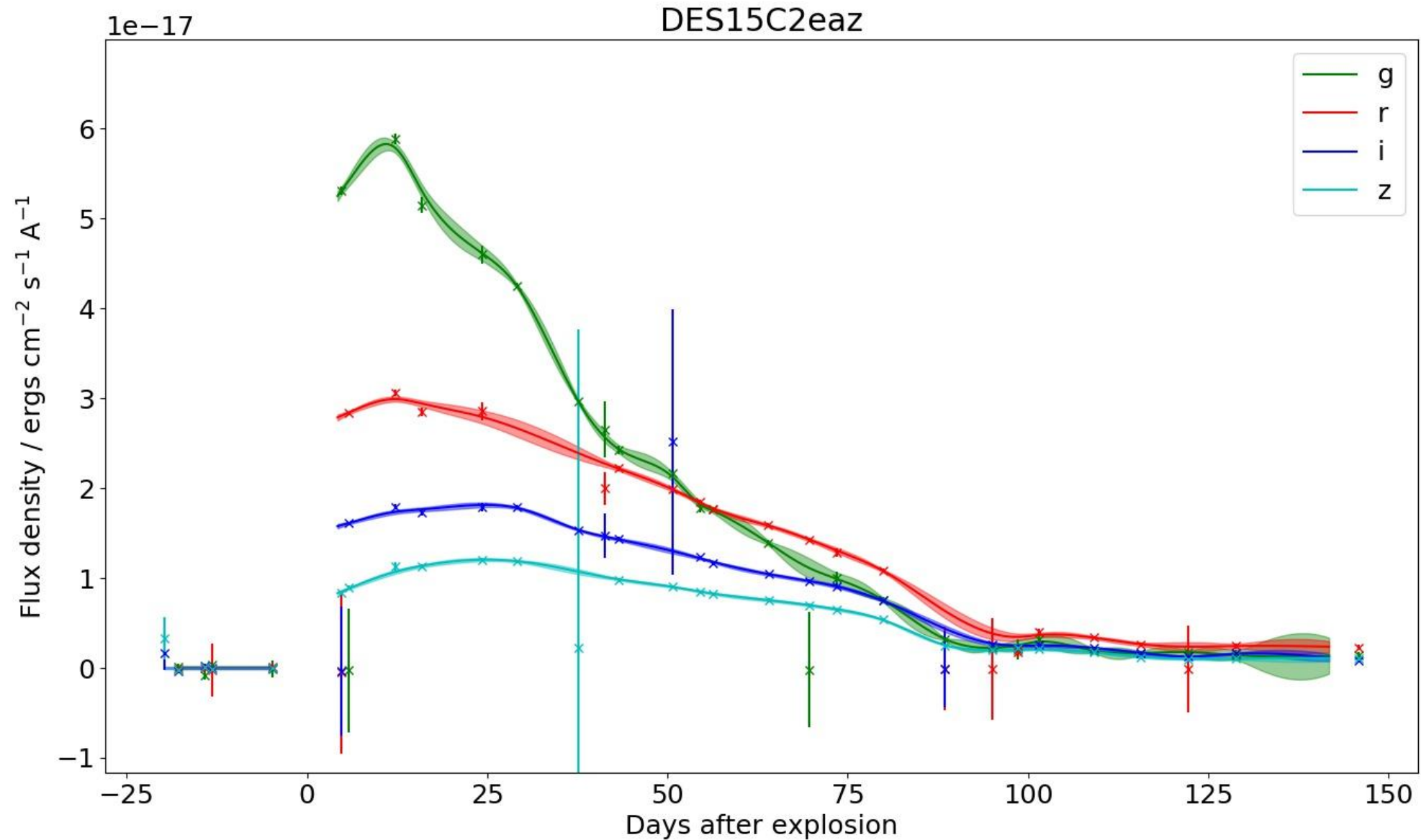
- 70 spectroscopically-confirmed CCSNe in DES, 46 after cuts
 - 37 Hydrogen-rich, 9 stripped-envelope



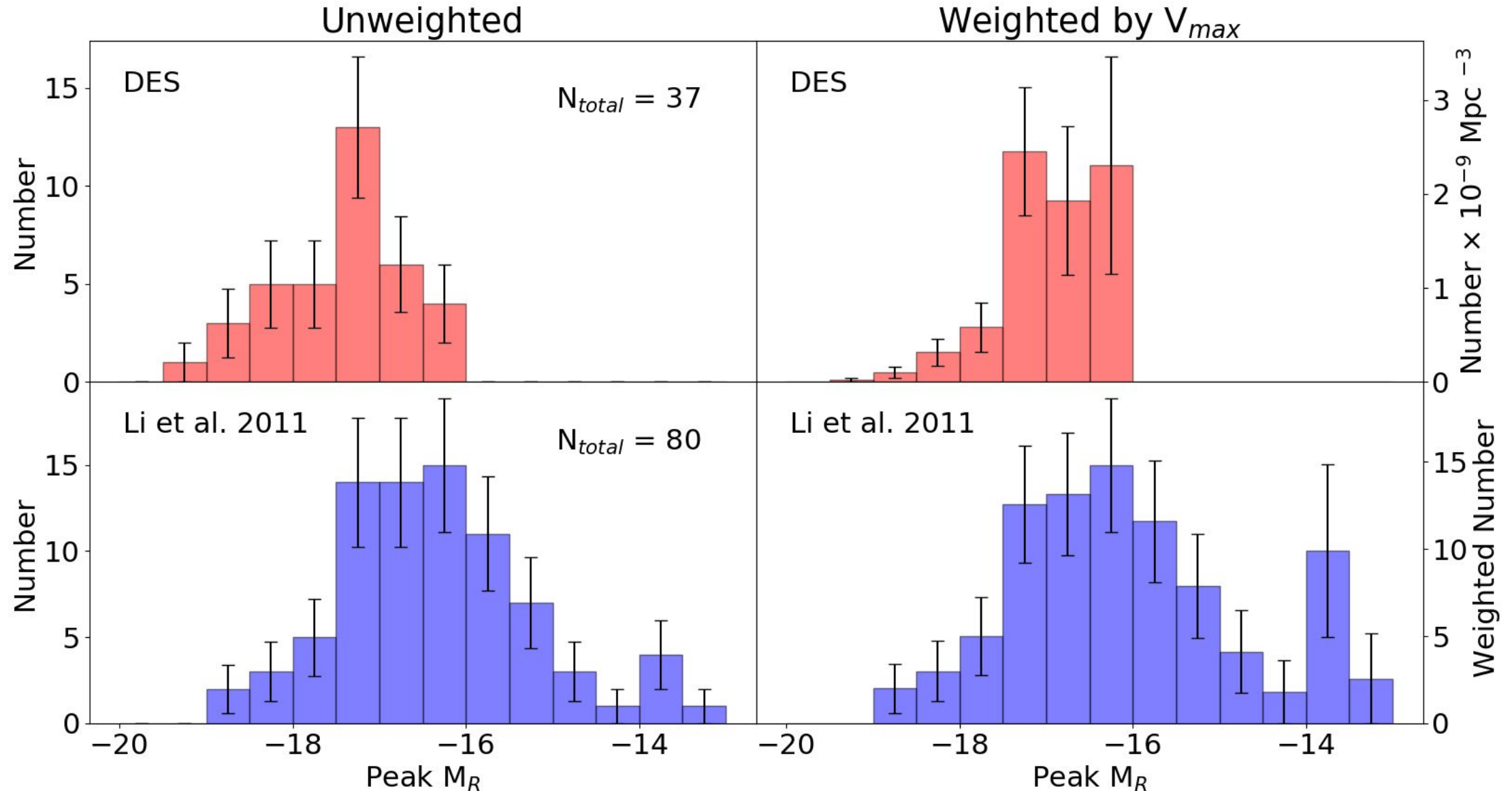
Analysis



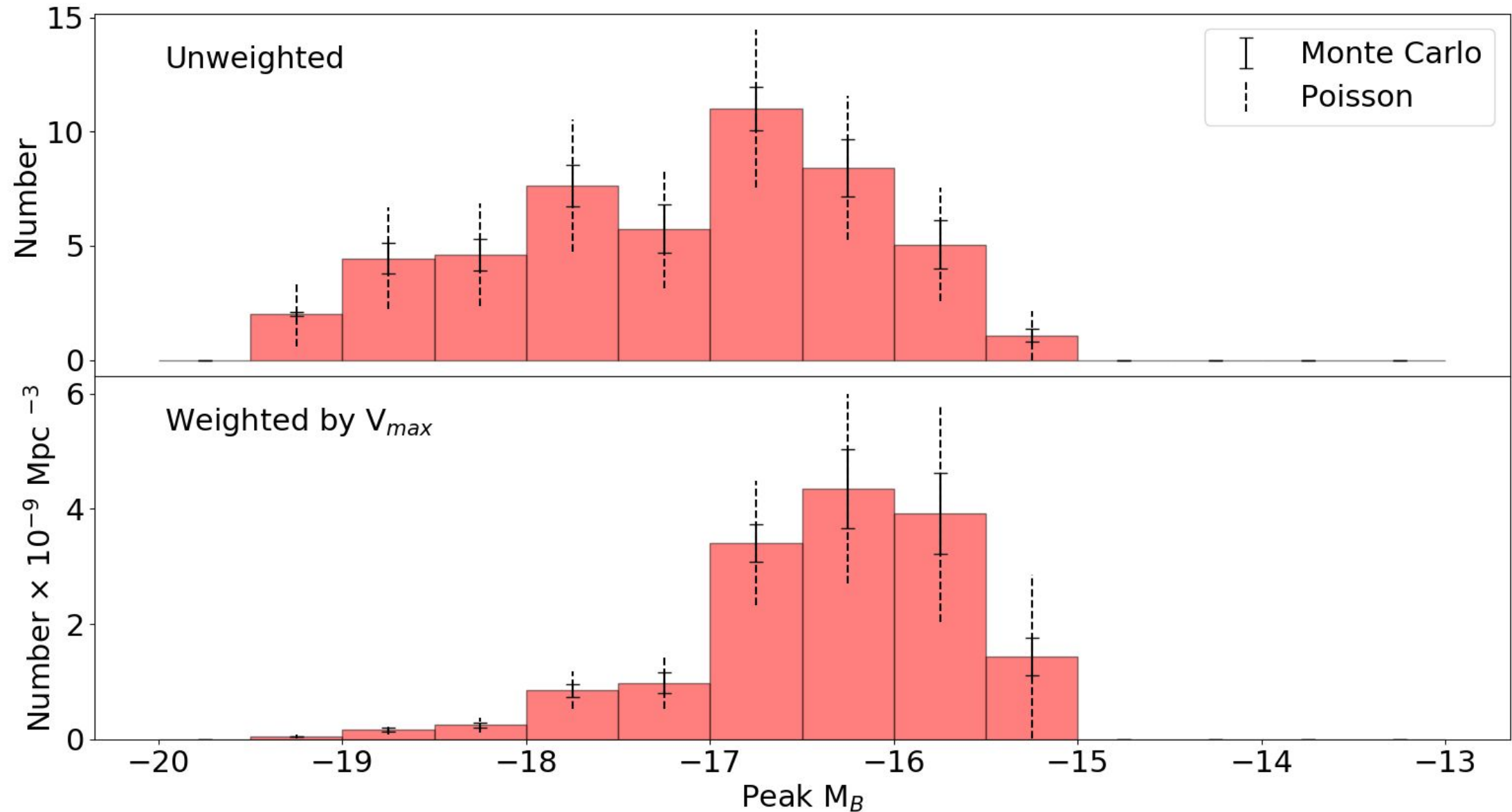
Analysis



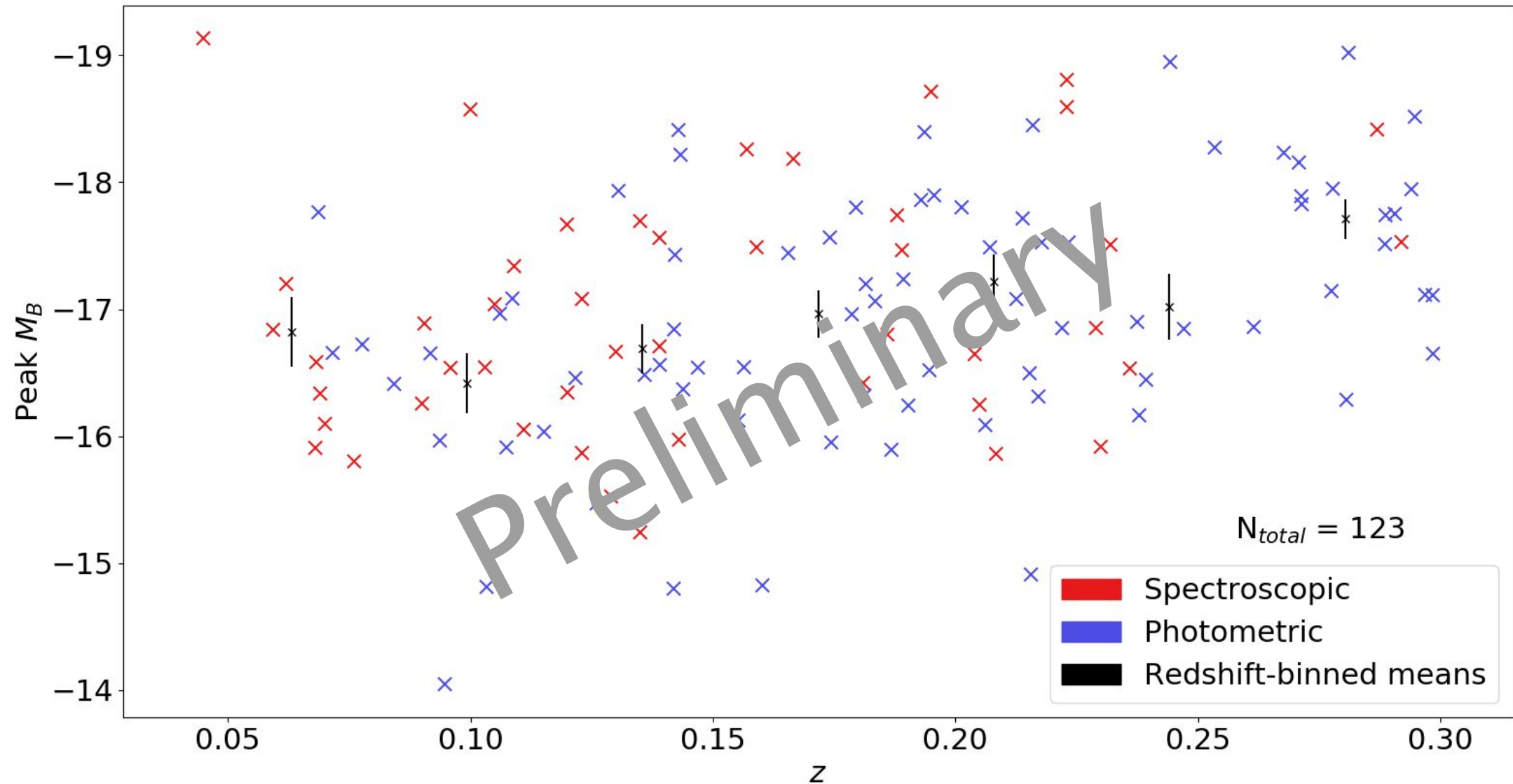
Results - Luminosity Functions (Spectroscopic SNe II - R band)



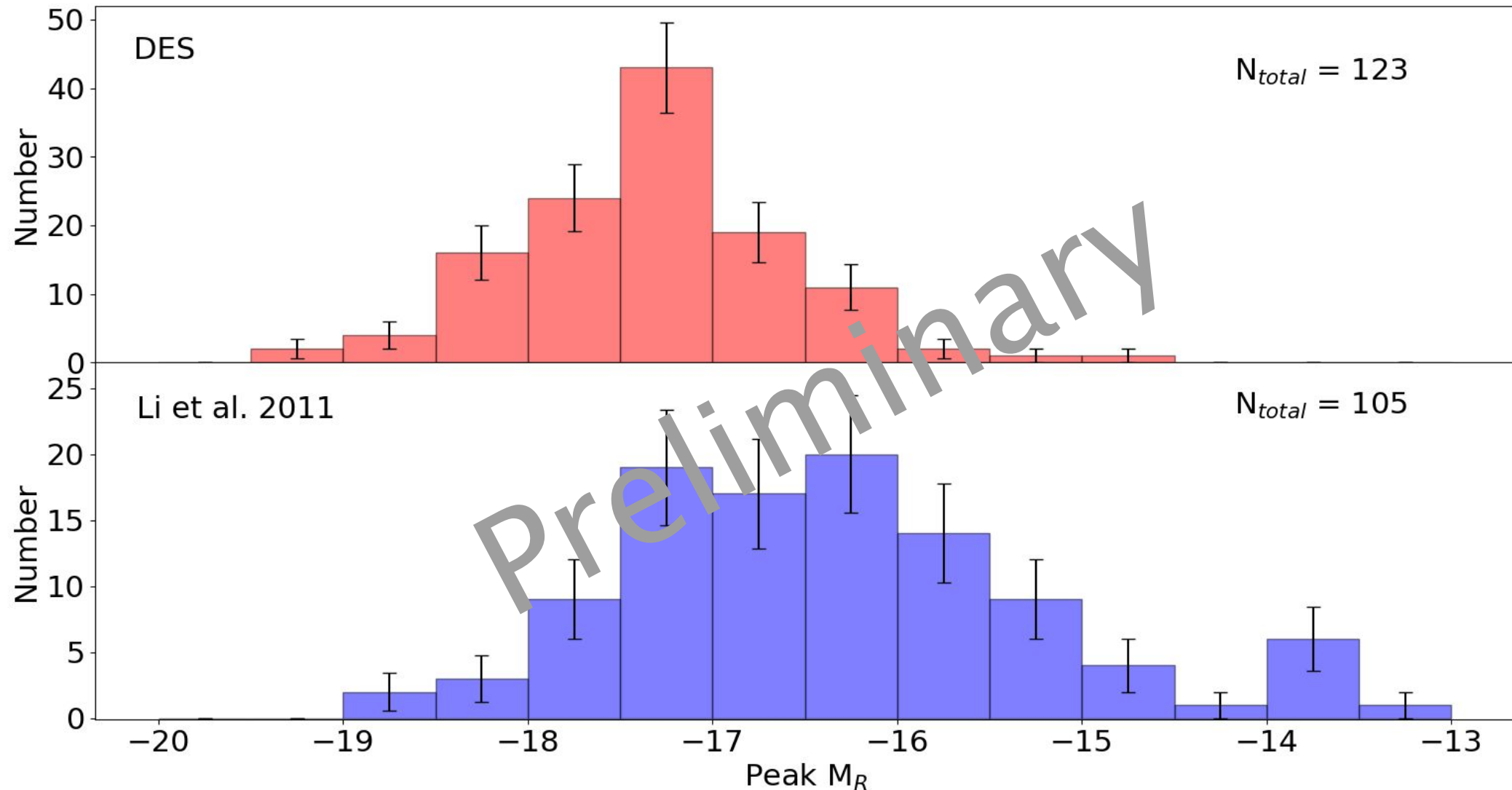
Results - Luminosity Functions (Spectroscopic SNe II - B band)



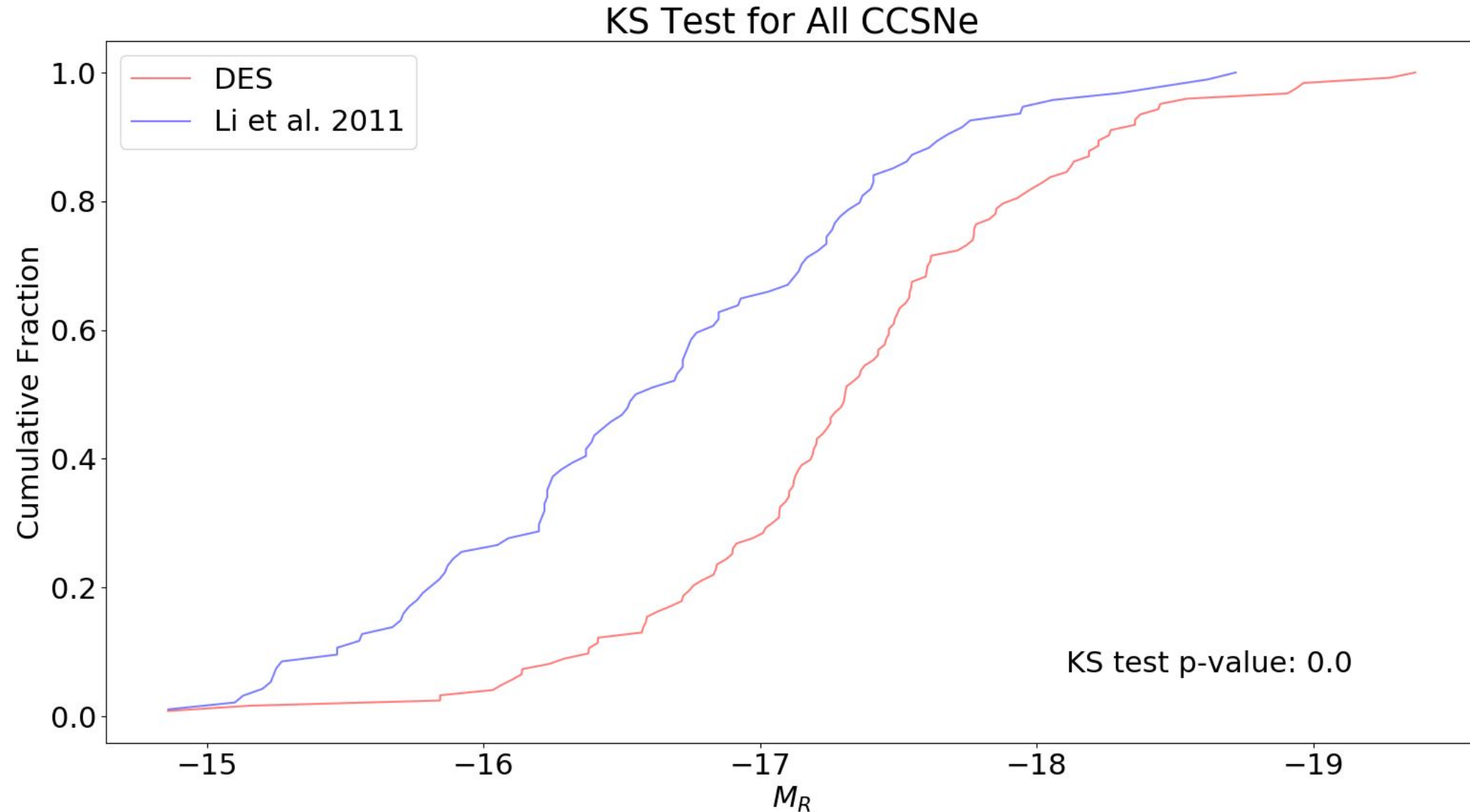
DES Sample - Spectroscopic and Photometric



Results - Luminosity Functions (All CCSNe including Photometric)



Results - KS Test (All CCSNe including Photometric)



Next steps

- Refine our photometric sample by looking into more sophisticated classification techniques (e.g. Recurrent Neural Networks)
- Look at an intermediate redshift sample to see if it appears to follow the same redshift evolution
- Look at the **hosts** of CCSNe DES